Student Industrial Project Presentation

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Content

Background and Project Statement

Strategic Mandate

- **I was attached to the BPI RU PMO (Internalization Team), the custodians of the "Systemic Shifts" strategic framework.**
- **Our Core Mission: To transform PETRONAS Upstream into a "Fitter, Focused, and Sharper" organization by driving strategic performance and internalizing this new culture across the workforce**

The Operational Observation

- While executing these initiatives, I managed the dissemination of information, resources and policy updates.
- The Reality: I observed that the foundational processes were heavily manual and disjointed. Story intake relied on static forms, drafting was offline, and design was disconnected from data sources

"It wasn't just a BPI team problem; it was a prominent issue across the entire PETRONAS Upstream business unit."

Fragmented & Manual Workflows

- Employees across Upstream are burdened with disjointed, manual processes for data handling and routine documentation
- This operational friction traps high-value talent in low-level administrative work, distracting them from focusing on strategic execution

Decentralized Institutional Knowledge

- Critical "Systemic Shifts" policies and frameworks are scattered across disparately organized repositories like SharePoint and email archives.
- Information retrieval becomes a difficult "hunt," creating tribal knowledge silos where consistent answers are hard to locate efficiently.

Rigid & Synchronous Knowledge Transfer

- Gaining deep understanding of complex strategies previously relied on scheduling formal classes or waiting for expert talks.
- There was no flexible, "on-demand" mechanism for staff to self-learn or verify their knowledge through accessible audio or testing formats.

Unscalable Creation & Analysis

- Strategic storytelling and asset management are hindered by slow manual drafting cycles and unstructured, unsearchable media libraries.
- The department faces significant production bottlenecks, unable to generate insights and content fast enough to sustain the cultural change momentum.

Problem Solution

Centralized Intelligence

Deploy Advanced RAG (Retrieval-Augmented Generation) to ground LLM answers in official PETRONAS policy documents and frameworks to eliminates the difficulty of retrieving scattered institutional knowledge by centralizing it into a single vector database, policies

Automated Workflow

Develop specialized Content and Meetings Agents to accelerate manual drafting cycles and documentation tasks to overcomes production bottlenecks caused by manual drafting to generate strategic stories and visual concepts

On-Demand Learning

Engineer Podcast and Quiz Agents to transform static knowledge into flexible, on-demand learning and assessment formats to breaks the reliance on synchronous training sessions.

Data & Visual Intelligence

Develop Visual and Analytics Agents to extract actionable insights instantly from unstructured images and raw datasets to unlocks value from visual assets and manual data processing to generate predictive insights.

VERA AI: AN AI ASSISTANT WITH ADVANCE RAG-POWERED INTELLIGENT KNOWLEDGE BASE AND LLM INTERGRATION TO ACCELERATE WORKFLOW OF PETRONAS UPSTREAM

VERA

An enterprise-grade intelligent assistant for PETRONAS Upstream, serving as the unified digital engine to operationalize strategic performance

The Ecosystem

VE tes a modular ecosystem of 6 specialized AI agents, integrated directly into the core platform to execute domain-specific operational tasks.

Name Origin

De atin "Veritas" (Truth), reflecting its core mandate to deliver strictly accurate, citation-backed answers rather than hallucinated responses.

Methodology & Approach

Hybrid Agile-SDLC for Enterprise AI | **Modern AI Development Lifecycle** | **End-to-End Hybrid AI Delivery Model**

~14 Weeks

Phase 1: Discovery & Planning

- Legacy Audit: Mapped existing manual workflows to identify knowledge silos.
- Data Taxonomy: Defined strict metadata standards for accurate Vector Database categorization.
- Scaffolding: Initialized "Serverless" architecture using Next.js 14 and Firebase.

Phase 2: Core Engineering

- RAG Pipeline: Built ingestion scripts with 800-token Semantic Chunking and text-embedding-3-large.
- Hybrid Compute: Bridged local GPU (PyTorch/CUDA) with Cloud Firestore for visual privacy.
- Logic Gate: Deployed stateless Cloud Functions with a 0.65 "Noise Gate" threshold to block hallucinations.

Phase 3: Agent Ecosystem

- Module Dev: Coded specialized logic for Analytics (Data Viz) and Meetings (JSON extraction) agents.
- UI Performance: Optimized React Server Components (RSC) to handle 6-agent loads seamlessly.
- Tuning: Refined "Adaptive Top-k" logic to expand retrieval when confidence drops below 0.78.

Phase 4: Validation & Security

- Stress Testing: Deployed "Adversarial Prompts" to verify the system rejects invalid queries 100% of the time.
- Production Readiness: Finalized Operational Runbooks and performed code splitting/lazy loading optimization to ensure the platform met enterprise performance standard.

Technology Stack

Frontend Layer

NEXT.JS

- **Next.js 14 (App Router):** Implements a Hybrid Rendering strategy using React Server Components (RSC) for fast initial loads (SEO) and Client Components for dynamic interactivity.

React

- **Interactive UI:** Integrated Framer Motion for GPU-accelerated animations and fluid layout transitions to enhance user engagement .

tailwindcss

- **Styling: Tailwind CSS configured with a centralized theme file to strictly enforce PETRONAS corporate branding (Teal/Emerald gradients) .**

Serverless Backend

node JS

- **Cloud Functions: Stateless Node.js middleware acting as a secure API Gateway to encapsulate API keys and handle business logic .**

Cloud Firestore

Firebase

- **Cloud Firestore: Real-time NoSQL database utilizing WebSocket listeners for sub-second UI synchronization of chat streams.**

Firebase Authentication

- **Security: Firebase Auth with Custom Claims to enforce strict Role-Based Access Control (RBAC)**

Hybrid AI Inference Engine

OpenRouter OpenAI

- **Text & Logic (Cloud): Integrates Google Gemini and OpenRouter for high-level reasoning and policy summarization.**

HUGGING FACE

PyTorch

- **Visual Synthesis (Local Edge):**
  - **Architecture: Custom Python 3.10 daemon running on PyTorch with CUDA hardware acceleration.**
  - **Optimization: Utilizes Hugging Face Diffusers with xFormers memory-efficient attention to optimize inference on consumer hardware.**

AI Deep Dive (The "Brain" of VERA)

Advanced RAG Architecture (Ingestion Layer)

Semantic Chunking

- Documents are not arbitrarily split; they are segmented into 800-token chunks with an 80-token overlap.
- This specific configuration was chosen to preserve contextual continuity across policy clauses while fitting multiple distinct sources into the LLM context window

High-Dimensional Embeddings:

- The system utilizes OpenAI's text-embedding-3-large model to transform text into 3,072-dimensional vectors.
- This high dimensionality was benchmarked against smaller models and selected for its superior ability to capture the nuanced technical jargon of the Oil & Gas industry .

Intelligent Retrieval Logic (The Decision Engine)

Noise Gate Threshold

- To eliminate hallucinations, the system enforces a strict Cosine Similarity threshold of 0.65.
- Any retrieved chunk scoring below this is mathematically rejected as "noise," triggering a fallback response ("No relevant document found") rather than fabricating an answer .

Adaptive Top-k Strategy:

- Standard State: Retrieves the top 5 most relevant chunks.
- Ambiguity Handling: If the top result's confidence score drops below 0.78, the system automatically expands retrieval to 8 chunks to cast a wider net for vague or complex user queries .

Context Engineering & Governance

Hierarchical Prompt Structure

Role Definition (Persona)

↓

Constraint Layer (Negative)

↓

Task/Format Layer (JSON/Markdown) 5.

XML Context Injection:

- Retrieved knowledge chunks are wrapped in XML tags (e.g., <source id="doc_1">) before injection.
- This allows the LLM to distinctly "see" document boundaries and generate precise, verifiable citations for every claim it makes

The 6 AI Agents (Reasoning & Function)

Analytics Agent

Hours spent cleaning data; trends missed due to volume.

↓

Ingests raw files to identify patterns, forecasts, and anomalies automatically.

Meetings Agent

Unstructured notes lead to forgotten tasks and items.

↓

Transcribes meeting notes into Executive Briefs and assigns owners to Action Items

Content Agent

Manual drafting takes days; struggles with brand alignment.

↓

Generates on-brand narratives and visuals in minutes using "precedent stories."

Visual Agent

Image silos are unsearchable; requires manual browsing.

↓

Uses Vision AI to auto-tag images, making the database fully searchable.

Podcast Agent

struggle to consume dense PDFs and Specific Topics

↓

Converts text policies into audio modules for "on-the-go" learning.

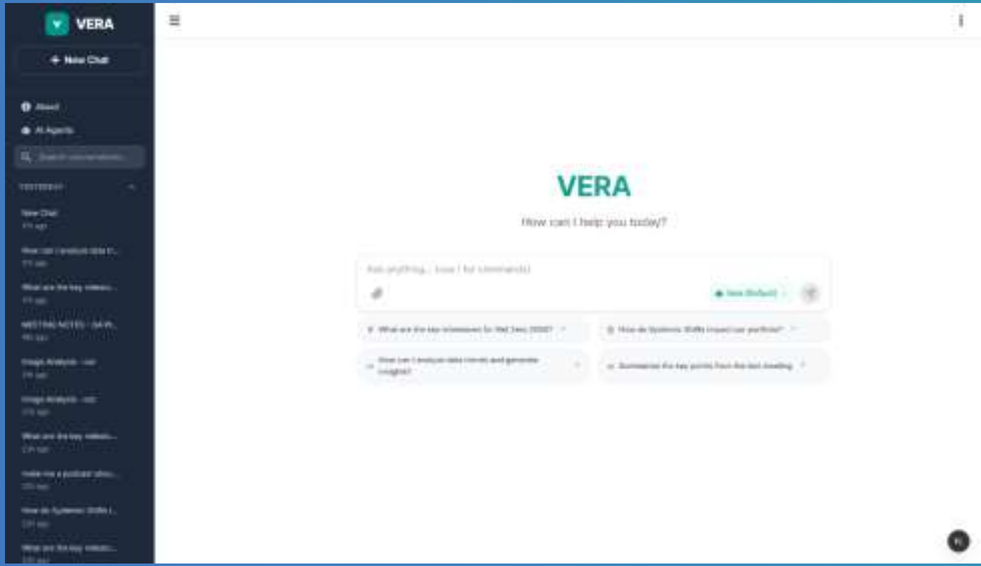
Quiz Agent

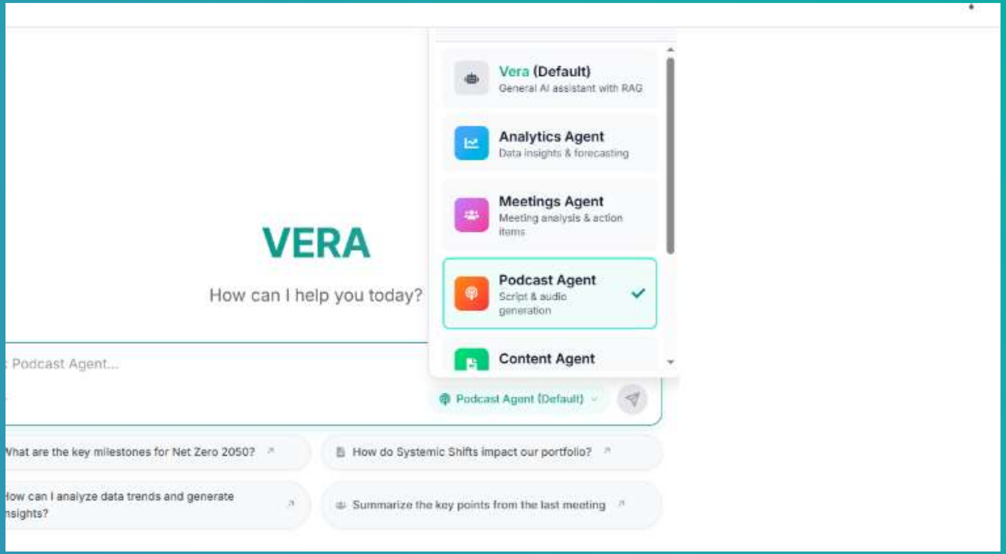
Creating assessments is a repetitive, manual task.

↓

Auto-generates quizzes directly from Knowledge Base materials.

Project Snippets and Demo





Project Snippets and Demo

“Live Demo”

Results & Discussion

Operational Efficiency

Drafting Velocity

- *Legacy Process: Manual data extraction and offline copywriting required ~1 Full Day of active work .*
- *VERA Workflow: Content Agent generates structured narrative and visual concept drafts within 5-15 Minutes (in ideal scenarios) .*

Knowledge Retrieval

- *Legacy Process: Staff spent 15-30 minutes navigating disparate SharePoint folders and email archives to locate specific policies .*
- *VERA Workflow: VERA AI retrieves citation-backed answers in 4.2 Seconds, enabling instantaneous access to institutional knowledge .*

Documentation

- *Legacy Process: Manual transcription and subjective extraction of action items, often leading to missed tasks or undefined owners .*
- *VERA Workflow: Meetings Agent automatically transcribes and structures Executive Briefs with assigned owners immediately .*

Quality & Trust Metrics:

Citation Integrity

Achieved a 94% Citation Accuracy rate, ensuring RAG responses are verifiable against official documents

User Acceptance:

Recorded an 82% First-Pass Acceptance Rate, confirming the AI output requires minimal human editing.

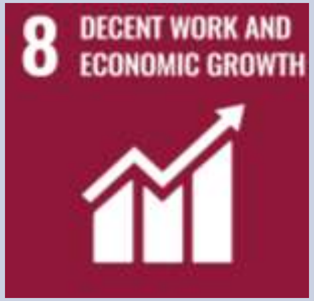
Hallucination Prevention:

100% Rejection Rate for out-of-domain queries (verified via the 0.65 Noise Gate threshold).

System Responsiveness:

Maintained <5 Second Latency for complex RAG queries, ensuring flow preservation.

Sustainability Impact (SDGs)



1 Decent Work & Economic Growth

- High-Value Repurposing: Shifts human talent from low-value manual data entry to high-level strategic execution.
- Mitigating Burnout: Automates repetitive administrative toil, improving work-life balance and reducing cognitive load.

2 Responsible Consumption & Production

- Sustainable Lifecycle: Dynamic indexing prevents data redundancy, ensuring existing knowledge is reused effectively rather than constantly recreated.
- Resource Efficiency: Maximizes the utility of existing digital assets before creating new ones.

3 Climate Action (Green AI)

- Reduced Energy Intensity: Drastically lowers active computing time by compressing drafting workflows from days to minutes.
- Carbon Footprint: Minimizes total device "active time" per task, aligning with Green AI and Net Zero Carbon Emissions (NZCE) 2050 goals.

Conclusion & Future Recommendations

Operationalizing the Mandate

❖ **Mission Accomplished:**

VERA AI successfully provided the digital infrastructure required to internalize the "Systemic Shifts" culture, shifting PETRONAS Upstream from manual, fragmented processes to a unified, intelligent ecosystem.

❖ **Barriers Removed:**

Effectively eliminated the "Search Tax" of tribal knowledge (P2) and the "Manual Toil" of drafting (P1), validating that AI can solve Upstream-specific operational bottlenecks.

❖ **Architectural Validation:**

Proved that a Hybrid Architecture (Advanced RAG for truth + Specialized Agents for workflows) is a secure, scalable blueprint for the department.

Operationalizing the Mandate

Enterprise Integration & Security

Azure AD SSO: Transition from standalone auth to Enterprise Single Sign-On to enable seamless user-based logging and secure collaboration.

User-Centric Data: Enable specific user bases to share meeting analysis and analytics dashboards securely between teams.

Deep Reasoning & Cognitive Scale

Knowledge Graphs: Augment Vector Search with Knowledge Graphs to map complex relationships (e.g., connecting specific Assets to Partners and Policies).

Expanded Context: Upgrade from free-tier limits to High-Token Models, enabling the processing of massive technical reports and complex reasoning tasks.

Next-Gen Agent Capabilities

Visual & Content: Integrate advanced generative APIs (e.g., Flux or Midjourney) for high-fidelity creative output.

Audio & Learning: Upgrade Podcast Agent to hyper-realistic neural voices with distinct styles (e.g., "Lecture Mode" vs. "Conversation")

Multimodal Input: Enable direct Voice Input for the Meeting Agent and complex, intricate data visualization for the Analytics Agent.

PETRONAS

50 YEARS